MILITARY SPECIFICATION

PENDANT

This specification is approved for use by the USA Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers polyurethane-elastomer-coated, double-braided-nylon pendants for use in the underwater, riser-portion of a mooring leg.

1.2. Classification. The polyurethane-elastomer-coated, double-braided-nylon pendants shall be 6-inch circumference before coating, with an eye and a thimble on each end.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

FSC 3835

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-T-23326	- Thimble, Plastic Rope.
MIL-R-24050	- Rope, Nylon, Double-Braided.

STANDARDS

FEDERAL

FED. STD. No. 601	- Rubber: Sampling and Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

DRAWINGS

ME

13218E8607	- Pendant.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SAMSON OCEAN SYSTEMS, INC.

Splicing Manual

(Application for copies should be addressed to the Samson Ocean Systems, Inc., 99 High Street, Boston, MA 02110.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D412 - Tension Testing of Vulcanized Rubber
D2240 - Indentation Hardness of Rubber and Plastics by Means of a Durometer
D750 - Operating Light and Weather Exposure Apparatus (Carbon-Arc Type) for Artificial Weather Testing of Rubber Compounds
D572 - Accelerated Aging of Vulcanized Rubber by Oxygen-Pressure Method

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Description. The polyurethane-elastomer-coated, double-braided-nylon pendants (hereinafter called the pendant) shall consist of a length of double-braided nylon rope with an eye spliced over a metal thimble at each end. The pendants shall be coated with polyurethane over their entire surface area to provide protection from abrasion and fish bite. Dimensions of the pendant shall be in accordance with Drawing 13218E8607.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the pendant. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more pendants as specified (see 6.2), for examination and testing within the time frame specified to prove prior to starting production that his production methods will produce pendants that comply with the require-

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ments of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government. When specified, the Government will conduct any or all of the preproduction examination and tests, as specified by the contracting officer (see 6.2).

3.3 Initial production. When specified by the contracting officer (see 6.2), the contractor shall furnish to the Government one or more pendants for inspection as specified in 4.4.

3.4 Material. Materials shall be as specified herein (see 6.8).

3.4.1 Nylon rope. Pendants shall be fabricated from nylon rope conforming to MIL-R-24050.

3.4.2 Thimbles. Thimbles shall conform to MIL-T-23326.

3.4.3 Polyurethane coating. The polyurethane elastomer coating shall meet the following requirements when tested as specified in 4.6.2.1:

Property	Requirement
Initial tensile strength, psi (min)	4,000
Ultimate elongation, % (min)	300
Tensile strength after immersion in distilled water at 160° F ± 2° F for 14 days, psi (min)	1,800
Tensile strength after immersion in 3% saltwater (by weight) at 160° F ± 2° F for 14 days, initial tensile strength retained, % (min)	40
Hardness, Shore Type A durometer, on specimen of at least .250-inch thickness, points (max)	90
Tensile strength after exposure to light for 100 hours, initial tensile strength retained, % (min)	75
Tensile strength after exposure to oxygen, initial tensile retained, % (min)	75

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3.5 Environmental conditions.

3.5.1 Operation. The pendants shall be capable of continuous immersion in saltwater at a pressure of 100 psi for a period of at least 2 years.

3.5.2 Storage. The pendants shall be capable of withstanding open storage for a period of not less than 8 years in a temperature range of 120° F to -25° F.

3.6 Detailing of technical requirements.

3.6.1 Splicing. Splicing used to made the eyes in both ends of the pendant shall be in accordance with Samson "Standard" Navy procedures as detailed in the Samson Splicing Manual. Splices shall have a strength not less than 85 percent of the new rope strength without slipping after one load application when tested in accordance with 4.6.4.

3.6.2 Penetration and thickness of coating. The coating shall penetrate into the nylon rope not more than 1/8 inch. The coating buildup on the outside surface of the nylon rope shall be sufficient to cover and protect all outer rope fibers, but not more than .093 inch except as specified in 3.6.3. The coating shall be uniform in thickness, smooth, and free from slits, cracks, and holes.

3.6.3 Eyes. The thickness of the coating applied to the inside portion of the thimbles shall be .010 to .031 inch.

3.7 Breaking strength. The breaking strength of the pendant shall be not less than 85 percent of the new rope strength when tested in accordance with 4.6.4.

3.8 Identification marking. The pendant shall be identified in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 End Item Final Inspection Requirement (EIFIR). Each pendant shall be inspected in accordance with the applicable EIFIR. One copy of the EIFIR shall be shipped with each inspected pendant (see 6.4.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction pendant shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction pendant shall be tested as specified in 4.6.4. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production pendants will be selected at random by the Government from the pendants being produced and will be examined as specified in 4.6.1 and tested as specified in 4.6.4 and 4.6.5 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production pendant shall not exclude the remaining pendants from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production pendant to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production pendant and shall be cause for refusal by the Government to continue acceptance of production pendants until evidence has been provided by the contractor that corrective action has been taken to eliminate

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the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pendants previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all pendants accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level II. Each sample shall consist of a 2-foot length of coated pendant.

4.5.2 Examination.

4.5.2.1 Individual. Each pendant shall be examined for major defects specified in 4.6.1. Presence of a major defect shall be cause for rejection.

4.5.2.2 Sampling. Samples selected in accordance with 4.5.1 shall be examined for the major and minor defects specified in 4.6.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5.3 Tests.

4.5.3.1 Individual. Each pendant shall be tested as specified in 4.6.4 and 4.6.5. Failure of any test shall be cause for rejection.

4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be tested as specified in 4.6.3.

4.6 Inspection procedure.

4.6.1 Examination. The pendant shall be examined as specified herein for the following defects:

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Number	Defect	Requirement Paragraph
Major		
101.	Dimensions not as specified.	3.1
102.	Material not as specified.	3.4
103.	Slits, cracks, or holes in coating.	3.4
104.	Nylon rope not as specified.	3.4.1
105.	Thimbles not as specified.	3.4.2
106.	Splicing not as specified.	3.7.1
Minor		
201.	Identification marking missing, incomplete, or not legible.	3.9

4.6.2 Tests.

4.6.2.1 Coating. The properties of the coating shall be tested as follows:

Property	Test Method
Initial tensile strength, psi (min)	FED. STD. No. 601, 4111
Ultimate elongation, % (min)	ASTM D412
Tensile strength after immersion in distilled water at 160° F ± 2° F for 14 days, psi (min)	ASTM D412
Tensile strength after immersion in 3% saltwater (by weight) at 160° F ± 2° F for 14 days, initial tensile strength retained, % (min)	FED. STD. No. 601, 6111 (In accordance with para. 4.8.1 of 6111)
Hardness, Shore Type A durometer, on specimen of at least .250-inch thickness, points (max)	ASTM D2240
Tensile strength after exposure to light for 100 hours, initial tensile strength retained, % (min)	ASTM D750
Tensile strength after exposure to oxygen, initial tensile strength retained, % (min)	ASTM D572

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4.6.2.2 Failure criteria. Failure to meet requirements specified in 3.4.3 shall constitute failure of these tests.

4.6.3 Samples. Each sample shall consist of 2-foot length of coated pendant. The thickness and penetration of the coating of each sample shall be measured in accordance with procedures specified by the contracting officer.

4.6.3.1 Failure criteria. Nonconformance to 3.6.2 shall constitute failure of this test.

4.6.4 Proof load. The pendant shall be tested as specified in 4.6.4.1 through 4.6.4.3. Presence of one or more defects shall be cause for rejection.

4.6.4.1 Working load test. The pendant shall be subjected to a tension load of 10,000 pounds. The tension shall be held for 10 minutes. At the end of the 10 minute period the pendant shall be examined for cracking of the polyurethane coating and other damage.

4.6.4.1.1 Failure criteria. Nonconformance to 3.6.1 and cracking of the coating or other visable damage shall constitute failure of this test.

4.6.4.2 Proof load test. The pendant shall be subject to a tension load of 25,000 pounds. The tension shall be held for 10 minutes then released and the pendant examined.

4.6.4.2.1 Failure criteria. Nonconformance to 3.6.1 and any evidence of broken strands or separation of the polyurethane coating from the pendant shall constitute failure of this test.

4.6.4.3 Breaking load test. The pendant shall be subjected to a proof tension load of 75,000 pounds. The proof load shall be held for 10 minutes.

4.6.4.3.1 Failure criteria. Failure of the pendant (breakage or separation of the eyes) shall constitute failure of this test.

4.7 Inspection comparison. The Government may select pendants at any time during the contract production period and subject these pendants to the examination specified in 4.5.2 and the tests specified in 4.5.3 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on pendants selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production pendant. In addition to any

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test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison pendant to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison pendant and shall be cause for refusal by the Government to continue acceptance of production pendants until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pendants previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all pendants accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Pendants not coiled and the coil secured as specified.
108.	Containers not as specified.
109.	Strapping not as specified.
110.	Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

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5.1.1 Level A. Each pendant shall be coiled in a minimum size coil and the coil secured. Each coiled pendant shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.1.2 Commercial. Pendants shall be packed to assure carrier acceptance and delivery to destination, without damage, at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking.

5.2.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.2.2 Commercial packaging. The marking shall reflect as a minimum, the item name, National Stock Number, quantity, contract number, and appropriate address markings. The marking shall be applied with a waterproof material, be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The pendant is intended for use as part of a mooring leg deployment device which is a component of an air transportable tactical tanker mooring and off-loading pipeline system for across-the-beach delivery of bulk fuel.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model and number of pendants and samples required (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and test shall be conducted by the contractor (see 3.2).
(d) When initial production inspection is required and the number of pendants and samples to be furnished (see 3.3).
(e) Degree of packing required (see 5.1).
(f) When other than Finish B strapping is required (see 5.1.1).
(g) Method of measuring and testing pendants (see 4.6.3 and 4.6.4).

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6.3 Preproduction model. Any changes or deviations of production pendants from the approved preproduction model during preproduction will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish pendants conforming to this specification.

6.4 Definitions. The following definition shall apply throughout this specification.

6.4.1 End Item Final Inspection Requirement (EIFIR). An EIFIR is a checklist of quality characteristics prepared by the Government which the contractor must follow during his final inspection of the completed end item to verify complete and functional conformance to contract requirements prior to submission for final acceptance (see 4.1.2).

6.5 Measuring and testing procedures. The contracting officer should specify methods of measuring and testing pendants.

6.6 Data requirements. The contracting officer should include requirements for technical publications pertaining to handling and storage of the pendant.

6.7 Provisioning. The contracting officer should include provisioning requirements for replacement pendants and instructions regarding shipment of pendants.

6.8 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodian:
Army - ME

User activities:
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Preparing activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-52945(ME) Pendant

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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